

RouteWise AI

Intelligent Supply-Chain Disruption Copilot & Fleet Resilience Control Tower

Track: AI	Event: IBM Bob AI Innovation Hackathon 2026
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The Challenge: Fragmented Visibility & Cold-Chain Fragility

Global supply chains suffer from severe information silos during unforeseen operational shocks:

Fragmented Disruption Telemetry	Compounding Cold-Chain Risks	Manual Decision Latency
Weather anomalies, port strikes, and canal blockages are reported across disconnected web feeds. Dispatchers lack unified spatial correlation to determine which shipments are directly impacted.	High-value pharmaceuticals (vaccines, biologics) require strict 2°C–8°C compliance. Unnoticed transit bottlenecks cause container battery drains and catastrophic cargo spoilage.	Evaluating alternative transshipment corridors, carrier reliability, freight surcharges, and fleet availability takes hours of manual spreadsheet reconciliation.

Operational Impact: Millions in cargo value exposed, regulatory compliance breaches, and low fleet utilisation during critical disruption spikes.

The Solution: RouteWise AI Intelligent Control Tower

RouteWise AI connects every stage of the logistics decision loop into an autonomous, proactive workflow:



Key Breakthroughs:

- **Deterministic Zero-Hallucination Risk Engine:** Standardized 0–100 scoring with human-auditable risk drivers.
- **Predictive Thermal Slope:** Alerts dispatchers *before* upper thresholds (8.0°C) are breached.
- **Conversational IBM Bob Copilot:** Natural language dialogue with full contextual database grounding.

Technical Architecture: Multi-Tier Resilient System

Layer	Technologies	Role & Responsibilities
Frontend Control Tower	React 19, TypeScript, Vite, Tailwind v4, Leaflet	Interactive live map, IoT temperature simulator, 1-click approval modal, IBM Bob conversational copilot drawer.
Companion Dashboard	Streamlit, Pandas, Python 3.13	Lightweight demonstration companion dashboard for instant single-command evaluation.
Backend Gateway	FastAPI, SQLAlchemy 2.0, Pydantic v2, SQLite / MySQL	High-speed asynchronous REST API, JWT authentication, CORS security, transaction management.
10 Decision Engines	Python Math, Spatial Correlators, Slope Analyzers	Disruption, Risk (0-100), Alternative Route, Carrier Ranking, Fleet Utilisation, Cold-Chain IoT, Alert & Audit.
MCP Protocol Server	JSON-RPC 2.0, Model Context Protocol Schema	Standardized AI tool integration: exposes active shipments, disruptions, route evaluation, and idle fleet queries.
AI / IBM Bob Layer	IBM WatsonX / Bob API + Deterministic Fallback	Natural language synthesis with strict factual grounding. Operates 100% offline if credentials are omitted.

Core Calculation & Surveillance Engines

1. Cold-Chain IoT Engine	2. Deterministic Risk Engine
• Configurable compliance profiles (e.g. Pharma 2°C–8°C) • Severity classification: NORMAL, WARNING, MAJOR, CRITICAL • Predictive thermal slope: calculates rate of change $\Delta T/\Delta t$ to forecast breach time before cargo is damaged.	• Standardized 0–100 scoring (Low, Medium, High, Critical) • Factors disruption proximity, delay hours, cargo value exposure, priority class, and active thermal excursions.
3. Multimodal Reroute Engine	4. Fleet Utilisation Engine
• Auto-generates bypass transshipment corridors • Evaluates delta transit hours, freight surcharges, and projected risk drop (e.g. 99/100 → 24/100 via Colombo) • Evaluates carrier performance & reefer certifications.	• Calculates exact network utilisation (Active vs Total Capacity) • Dwell-time monitoring: detects idle assets (e.g. TRUCK-205 in Ahmedabad idle 8h) • Recommends repositioning to surge bottleneck corridors.

IBM Technology Stack & Bob Copilot Integration

RouteWise AI incorporates IBM core technologies across the intelligence pipeline:

Technology	Implementation Architecture	Operational Benefit
IBM Bob Conversational Copilot	Dedicated conversational drawer and REST endpoint (/api/ai/chat) powered by supply-chain intent routing.	Logistics managers query network health naturally ('Which shipments are most urgent?', 'Should SH-1024 be rerouted?') with instant evidence.
IBM watsonx.ai / Granite	Targeted prompt engineering with strict operational grounding context and hyperparameter tuning.	Synthesizes clear situation overviews, risk rationale, and tradeoff comparisons without fabricating external data.
Model Context Protocol (MCP)	Standardized tool registry exposing 6 core logistics functions via JSON-RPC stdio and REST.	Enables external AI agents, IDE assistants, and enterprise systems to query telemetry, disruptions, and rerouting tools.
Graceful Offline Fallback	Deterministic local decision support engine activates if IBM credentials are unconfigured.	Guarantees 100% operational uptime for mission-critical supply chains during air-gapped or offline scenarios.

Verification & 1-Click Hackathon Demonstration Flow

Demonstrated End-to-End Scenario:

- 1. **Baseline State:** 53 active shipments, 3 disruptions, 68.8% fleet utilisation.
- 2. **Disruption Event:** Mumbai Port Strike (High Severity, 4-day closure). SH-1024 (Vaccines, \$620,000, Mumbai → Rotterdam) origin is blocked.
- 3. **IoT Excursion:** Simulated thermal rise (7.8°C → 8.4°C). Predictive slope alerts imminent breach of upper 8.0°C compliance bound. Risk spikes to **99/100 (CRITICAL)**.
- 4. **Autonomous Recommendation:** Reroute via *Alternative A (Colombo Transshipment Hub)* (+18h, +\$8,400, risk drops to 24) and redeploy idle *TRUCK-205* from Ahmedabad to relieve backlog.
- 5. **1-Click Action & Audit:** Manager approves reroute; decision recorded in immutable audit log; fleet utilisation rises to 72.2%.

Automated Test Suite Status	Result
Pytest Backend Unit Test Suite (19 test cases)	19 / 19 PASSED (100%)
Live End-to-End API Integration Suite (15 test workflows)	15 / 15 PASSED (100%)
Model Context Protocol (MCP) Tool Verification	6 / 6 TOOLS VERIFIED
React 19 + TypeScript + Vite Frontend Build	0 ERRORS (618ms Build)

Team CHARUSAT Innovators & Future Roadmap

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Future Development Roadmap:

- **Live Telematics Ingestion:** Connect real-time GPS & OBD-II telemetry streams via MQTT/Kafka.
- **Autonomous Carrier Bidding:** Multi-agent negotiation protocol for automated spot rate booking during emergency reroutes.
- **Predictive Micro-Climate Modeling:** Ingest NOAA and ECMWF global weather forecast grids into spatial risk matrices.
- **Enterprise ERP Bridges:** Turnkey bi-directional integrations with SAP Transportation Management and Oracle SCM Cloud.